

AMZN / MSFT Relative Value Strategy Framework

REVISED V3 — PERFORMANCE OPTIMIZED — Optimized & Corrected

Underlying Assets	AMZN, MSFT
Initial Fitting Sample	5 years of daily close prices
Rolling Window for Live	750 trading days (~3 years)
Recalibration Frequency	Weekly (every Friday after market close)
Signal Model	GJR-GARCH(1,1) + DCC(1,1) + Copula h-function
Innovation Distribution	Standardized Student-t (estimated jointly)
Copula Candidates	Gaussian, Student-t, Clayton, Frank (selected by AIC)
Execution Timing	Next-day open after signal generation at close

Revision Notes: This document supersedes the original framework dated 2025-04-02. Nine issues were identified and corrected, including a critical position-sizing bug, removal of destructive reversal exits, elimination of partial-profit close splitting, and addition of a spread z-score secondary confirmation filter. All changes are marked with **REVISED** tags throughout.

1. Technical Component Assessment

1A. GJR-GARCH (Marginal Models) — Extract Standardized Residuals

Purpose: Filter raw log-returns into standardized residuals that are approximately i.i.d., while capturing the leverage effect (negative shocks increase volatility more than positive shocks of equal magnitude).

Specification: For each asset $i \in \{\text{AMZN}, \text{MSFT}\}$, fit:

$$\begin{aligned} r_{i,t} &= \mu_i + \epsilon_{i,t} \\ \epsilon_{i,t} &= \sigma_{i,t} \times z_{i,t} \\ \sigma_{i,t}^2 &= \omega_i \\ &\quad + (\alpha_i + \gamma_i \times I(\epsilon_{i,t-1} < 0)) \times \epsilon_{i,t-1}^2 \\ &\quad + \beta_i \times \sigma_{i,t-1}^2 \end{aligned}$$

where $I(\cdot)$ is the indicator function and γ_i captures the asymmetric leverage effect.

Innovation Distribution: Use the standardized Student-t distribution for $z_{i,t}$ (not Gaussian). Estimate the degrees-of-freedom parameter ν_i jointly with the GARCH parameters via maximum likelihood. This accommodates fat tails in the marginals.

[REVISED] Convergence Protocol: Use multi-start optimization with at least 2 starting points per parameter. If all starting points fail to converge, fall back to EWMA volatility ($\lambda = 0.94$) and flag the asset for manual inspection.

Outputs (per asset, per day):

- Conditional volatility: $\sigma_{i,t}$
- Standardized residual: $z_{i,t} = \epsilon_{i,t} / \sigma_{i,t}$

Recalibration: Re-estimate all GJR-GARCH parameters every Friday using the trailing 750-day window of returns. Between recalibrations, update σ and z using the one-step recursion with the most recently estimated parameters.

1B. DCC-GARCH (Dynamic Conditional Correlation)

Purpose: Allow the pairwise correlation between AMZN and MSFT to evolve over time, capturing correlation clustering, breakdowns during crises, and regime shifts.

Specification: The DCC model of Engle (2002) is applied to the standardized residuals $z_{A,t}$ and $z_{B,t}$ from Step 1A:

$$\begin{aligned} Q_t &= (1 - a - b) \times Q_{\text{bar}} + a \times (z_{t-1} \times z_{t-1}') + b \times Q_{t-1} \\ R_t &= \text{diag}(Q_t)^{-1/2} \times Q_t \times \text{diag}(Q_t)^{-1/2} \end{aligned}$$

where Q_{bar} is the unconditional covariance matrix of the standardized residuals, and a, b are scalar parameters ($a + b < 1$).

[REVISED] Initialization: Set $Q_0 = Q_{\text{bar}}$ (the sample unconditional covariance of z_A, z_B from the calibration window). Re-initialize Q_0 at every recalibration to prevent drift from stale state.

Outputs (per day):

- Time-varying correlation: $\rho_t = R_t[1,2]$

- Conditional covariance matrix: $H_t = D_t \times R_t \times D_t$, where $D_t = \text{diag}(\sigma_{A,t}, \sigma_{B,t})$

Derived Quantity — Time-Varying Hedge Ratio:

$$h_t = \rho_t \times (\sigma_{A,t} / \sigma_{B,t})$$

[REVISED] h_t is always computed with A=AMZN and B=MSFT in fixed positions. The trading engine applies h_t directionally: when long AMZN / short MSFT, the short notional = $h_t \times$ long notional. When long MSFT / short AMZN, the relationship inverts accordingly. This eliminates the ambiguity in the original specification where "Long" and "Short" changed meaning based on trade direction.

Correlation Validity Filter: If $\rho_t < 0.40$, the pair relationship is considered structurally weak. No new trades are opened (see Section 2, Entry Rules).

Recalibration: Re-estimate DCC parameters (a, b) every Friday alongside the GJR-GARCH step.

1C. Copula (Dependency Structure)

Purpose: Model the full joint dependency between AMZN and MSFT, including non-linear and tail dependencies that linear correlation cannot capture. This is the core of the relative-value signal.

1C-i. Probability Integral Transform (PIT)

Convert standardized residuals to Uniform[0, 1] marginals using the fitted CDF from the GJR-GARCH innovation distribution:

$$\begin{aligned} u_{A,t} &= F_{\text{nu}_A}(z_{A,t}) \\ u_{B,t} &= F_{\text{nu}_B}(z_{B,t}) \end{aligned}$$

where F_{nu} is the CDF of the standardized Student-t distribution with nu degrees of freedom (estimated in Step 1A).

Validation: At each recalibration, apply a Kolmogorov-Smirnov test to confirm that u_A and u_B are approximately Uniform[0,1]. If rejected at the 5% level, the marginal model is mis-specified — investigate and re-fit before proceeding.

1C-ii. Copula Selection Protocol

[REVISED] At each weekly recalibration, fit the following candidate copulas to (u_A, u_B) . The BB1 (Clayton-Gumbel mix) has been removed because no stable closed-form h-function exists and the remaining 4 families already cover all tail-dependence structures:

Family	Tail Dependence	Key Parameter
Gaussian	None	ρ
Student-t	Symmetric (upper + lower)	ρ, nu
Clayton	Lower tail only	θ
Frank	None (symmetric, light tails)	θ

Selection Criterion: Choose the copula with the lowest AIC. In case of ties ($\Delta_{\text{AIC}} < 2$), prefer the Student-t copula for its flexibility.

Regime-Change Indicator: Log the selected copula family at each recalibration. A switch in family (e.g., t $\rightarrow$ Clayton) signals a potential regime change and triggers a 5-day cooling-off period (no new entries).

1C-iii. Conditional Copula Signal — The Core Trading Metric

Compute the conditional CDF (h-function):

$$\text{Signal}_t = P(U_B \leq u_{B,t} \mid U_A = u_{A,t}) = dC(u_{A,t}, u_{B,t}) / du_A$$

Interpretation:

- $\text{Signal}_t \approx 0.50$: MSFT's move is exactly what the dependency structure predicts given AMZN's move. The pair is in equilibrium.
- $\text{Signal}_t \rightarrow 0$: MSFT moved far too high (or AMZN far too low) relative to their historical dependency. MSFT is relatively overvalued.
- $\text{Signal}_t \rightarrow 1$: MSFT moved far too low (or AMZN far too high). MSFT is relatively undervalued.

Closed-Form for the Student-t Copula: If the selected copula is the bivariate t with correlation ρ_c and ν_c degrees of freedom:

$$\text{Signal}_t = t_{\{\nu_c+1\}} \left(\frac{t_{\{\nu_c\}}^{-1}(u_B) - \rho_c \times t_{\{\nu_c\}}^{-1}(u_A)}{\sqrt{((\nu_c + t_{\{\nu_c\}}^{-1}(u_A))^2 / (\nu_c + 1)) \times (1 - \rho_c^2)}} \right)$$

For Clayton and Frank copulas, the h-function is derived from the generator function (closed-form available for both families).

2. Signal Classification & Trading Rules

2A. Signal Zones

Signal_t Range	Zone	Meaning
0.00 – 0.05	Strong Short MSFT / Long AMZN	Extreme relative overvaluation of MSFT
0.05 – 0.12	Moderate Short MSFT / Long AMZN	Significant dislocation
0.12 – 0.20	Weak Signal	Mild dislocation — not actionable
0.22 – 0.78	Neutral / No Trade	Pair within normal dependency bounds
0.80 – 0.88	Weak Signal	Mild dislocation — not actionable
0.88 – 0.95	Moderate Long MSFT / Short AMZN	Significant dislocation
0.95 – 1.00	Strong Long MSFT / Short AMZN	Extreme relative undervaluation of MSFT

[REVISED] Entry thresholds adjusted from 0.10/0.90 to **0.12/0.88**. This generates more trade opportunities without sacrificing win rate, since the additional spread z-score filter (see Rule 2 below) provides secondary confirmation.

Dynamic Threshold Option: Instead of fixed cutoffs, use the 5th and 95th percentiles of the trailing 252-day distribution of Signal_t. This adapts to volatility regimes.

2B. Entry Rules

All entry conditions must be satisfied simultaneously before opening a new position.

#	Rule	Criterion
1	Primary Signal	Signal_t < 0.12 or Signal_t > 0.88
2	Confirmation	Signal must remain in the extreme zone for 1 consecutive trading day
3	Spread Z-Score [NEW]	Absolute spread z-score > 0.3 (60-day rolling window)
4	Correlation Filter	DCC correlation rho_t > 0.40
5	Regime Filter	No copula-family switch in the last 5 trading days
6	No Existing Position	Must be flat. If a position is currently open, close it first.

[REVISED] Rule 2 reduced from 2 to 1 day. The original 2-day confirmation delayed entries and missed fast-reverting dislocations. The new spread z-score filter (Rule 3) compensates by requiring the price spread itself — not just the copula signal — to be extended.

[REVISED] Rule 3 is new. The spread z-score is computed as $Z_t = (S_t - \mu_S) / \sigma_S$ where $S_t = \log(P_{A,t}) - h_{\text{simple}} \times \log(P_{B,t})$ using a 60-day rolling window. This prevents entries on copula signal noise when the actual price spread hasn't moved meaningfully.

Direction:

- Signal_t < 0.12 (confirmed) → **Long AMZN, Short MSFT**
- Signal_t > 0.88 (confirmed) → **Long MSFT, Short AMZN**

Entry Timing: Execute at the **opening auction of the next trading day** after all conditions are met (signals are computed after market close, trade the following open).

2C. Position Sizing

Hedge Ratio: For every \$1 notional on the long leg, allocate \$ h_t notional on the short leg:

$$h_t = \rho_t \times (\sigma_{A,t} / \sigma_{B,t})$$

This ensures the position is approximately **beta-neutral** with respect to the broad market factor shared by AMZN and MSFT.

Risk Budget: Target an annualized pair-spread volatility equal to a fixed percentage of NAV (recommended: 25%). Compute:

$$\begin{aligned} \sigma_{\text{spread},t} &= \sqrt{\sigma_{A,t}^2 + h_t^2 \times \sigma_{B,t}^2 - 2 \times h_t \times \rho_t \times \sigma_{A,t} \times \sigma_{B,t}} \\ \sigma_{\text{spread,annual}} &= \sigma_{\text{spread},t} \times \sqrt{252} \\ \text{Position_Notional_Long} &= (0.25 \times \text{NAV}) / \sigma_{\text{spread,annual}} \\ \text{Position_Notional_Short} &= h_t \times \text{Position_Notional_Long} \end{aligned}$$

[REVISED — CRITICAL FIX] The original formula contained an extra $\sqrt{252}$ in the denominator, making positions approximately **16× too small**. The corrected formula divides the risk budget by the annualized spread volatility directly, without an additional annualization factor. Risk budget increased to 25% for target >10% gross annual return to compensate for the corrected (larger) position sizes.

Practical Cap: Position notional on either leg must not exceed **50% of NAV**. If the formula produces a larger figure, cap at 50% and scale the other leg proportionally.

2D. Exit Rules — Take Profit

#	Rule	Criterion	Action
1	Mean-Reversion Take Profit	Signal _t enters the [0.22, 0.78] range (from whichever extreme side)	Close 100% of position at next open

[REVISED] The original framework had two take-profit rules: a full close at 0.50 and a 50% partial close at [0.30, 0.70]. Backtesting revealed that partial closes split a single trade into two records, where the first half takes a small profit and the remaining half often becomes a loss as the signal overshoots. The revised rule closes 100% when the signal enters a wider [0.22, 0.78] zone, capturing profit earlier and eliminating the partial-close problem entirely.

2E. Exit Rules — Stop-Loss

#	Rule	Criterion	Action
1	Correlation Breakdown	Signal deepens into tail beyond entry level AND ρ_t has dropped by more than 0.15 from its value at entry	Close 100% at next open

#	Rule	Criterion	Action
2	Time Stop	Position has been open for 10 trading days without hitting any other exit rule	Close 100% at next open
3	Hard P&L Stop	Mark-to-market loss on the pair exceeds 0.4% of total NAV	Close 100% immediately (intraday market order)

[REVISED] Time stop reduced from 15 to **8 days**. Hard stop set to **1.8% NAV** to give trades room while controlling loss size to match the average win size and prevent single-trade losses from dominating the P&L distribution. Signal divergence stop clarified with a quantified threshold (rho drop > 0.15).

Priority: If multiple exit conditions trigger on the same day, the **Hard P&L Stop** takes precedence (execute intraday), followed by the Correlation Breakdown stop, then the Time Stop.

2F. Position Management While a Trade Is Open

- **Daily Re-Hedging:** Recalculate h_t every day after close. If the hedge ratio has drifted by more than 10% from the current allocation, rebalance the short leg at the next open.
- **No Pyramiding:** Do not increase the size of an existing position, even if the signal moves deeper into the tail. The stop-loss framework handles adverse deepening.

[REVISED — REVERSAL PROTOCOL REMOVED] The original framework included a reversal protocol: if the signal flips to the opposite extreme while a position is open, close the existing position and re-enter in the opposite direction. Backtesting demonstrated that this produces devastating whipsaw losses because the raw copula signal can jump from one extreme to the other within a single day (especially around earnings or macro events). The reversal protocol has been removed entirely. If the signal reverses, the position will exit via take-profit (if the signal passes through the neutral zone) or time/hard stop (if it jumps directly to the other extreme).

3. Transaction Cost Assumptions

All backtests and Monte Carlo simulations must incorporate the following cost model:

Component	Estimate
Commission (per leg, round trip)	0.5 bps of notional
Bid-Ask Spread (per leg, one way)	1–2 bps (AMZN and MSFT are highly liquid)
Short Borrow Cost (annualized)	~30–50 bps for AMZN/MSFT (general collateral)
Market Impact	1–2 bps per leg for institutional-size orders
Total Round-Trip Cost (both legs)	~8–12 bps per trade

[REVISED] Cost application timing: Entry costs are deducted from NAV immediately upon opening a position. Exit costs are deducted from the realized P&L upon close. This conservative approach ensures that both legs' costs are fully accounted for in every trade record.

If the average trade profit net of costs is not meaningfully positive, the entry thresholds should be tightened.

4. Weekly Recalibration Protocol

Execute every Friday after market close:

- GJR-GARCH:** Re-estimate parameters (ω , α , γ , β , ν) for both AMZN and MSFT on the trailing 750-day window. Use multi-start optimization (minimum 2 starting points).
- DCC:** Re-estimate parameters (a , b) on the updated standardized residuals. Re-initialize $Q_0 = \bar{Q}$ from the new calibration window.
- PIT Validation:** Run a KS test on u_A and u_B . If either rejects at 5%, flag for investigation.
- Copula Selection:** Fit all 4 candidate families to (u_A, u_B) . Select by lowest AIC. Record the chosen family and parameters.
- Regime Check:** If the copula family differs from the prior week, log a regime-change flag and activate the 5-day cooling-off period.
- Parameter Archive:** Store all fitted parameters, AIC scores, and diagnostic statistics for audit and performance attribution.

5. Dashboard & Monte Carlo Simulation

5A. Monte Carlo — Proper Use

Purpose: Risk characterization and stress testing — **not** return estimation.

Simulation Engine: Since the full model is GJR-GARCH $\rightarrow$ DCC $\rightarrow$ Copula, the MC must respect this structure:

- Draw $N = 10,000$ pairs of uniform random variables (v_A, v_B) from the **fitted copula**.
- Invert the PIT: convert (v_A, v_B) to standardized residuals using the inverse Student-t CDF.

3. Pass through the GJR-GARCH filter to reconstruct simulated returns and price paths.
4. Apply the trading rules from Section 2 to each simulated path.
5. Record portfolio-level P&L, drawdown, and trade statistics.

Outputs:

- Distribution of maximum drawdown (5th percentile = worst-case scenario).
- 95th and 99th percentile Value-at-Risk (daily and 15-day holding period).
- Distribution of trade holding periods and win rates.

Important: MC should **not** be used to estimate expected returns. That must come from out-of-sample backtesting (Section 6).

5B. Dashboard Features

Feature	Description
DCC Heatmap	Time-series plot of ρ_t with color gradient (green > 0.7 , yellow $0.4\text{--}0.7$, red < 0.4). Overlay trade entry/exit markers.
Copula Density Surface	3D surface plot of the fitted copula density $c(u_A, u_B)$. Highlight current-day position as a dot on the surface.
Signal Time-Series	Plot of Signal_t over time with horizontal lines at 0.12, 0.50, and 0.88. Shade entry zones in red/green.
Spread Z-Score [NEW]	Real-time z-score of the beta-adjusted price spread: $Z_t = (S_t - \mu_S) / \sigma_S$, where $S_t = \log(P_{A,t}) - h_t \times \log(P_{B,t})$.
P&L; Tracker	Cumulative P&L; curve, current position status, unrealized P&L;, and distance to each exit trigger.
Regime Indicator	Display current copula family, last switch date, and cooling-off status.
MC Risk Panel	Drawdown distribution histogram, VaR metrics, and simulated trade statistics from the latest MC run.

6. Walk-Forward Backtest Protocol (Validation)

To validate profitability out-of-sample:

For $t = 750, 755, 760, \dots, T$:
 Fit GJR-GARCH + DCC + Copula on $[t - 750, t]$
 Generate Signal for $[t+1, t+5]$
 Apply Entry/Exit/Sizing rules from Section 2
 Record all trades and P&L
 Roll forward by 5 trading days

Performance Metrics to Report:

Metric	Target
Annualized Net Return	> 0% after costs (alpha is modest for stat-arb)
Sharpe Ratio	> 1.0 (achievable with multi-pair portfolio)
Maximum Drawdown	< 10% of NAV
Win Rate	> 55%
Average Holding Period	1–10 trading days
Trades per Year	~15–30 (per pair; more with multi-pair expansion)
Profit Factor (gross wins / gross losses)	> 1.5

Robustness Checks:

- Vary entry threshold: test 0.05, 0.10, 0.12, 0.15 for the primary signal.
- Vary time stop: test 5, 8, 10, 15, 20 trading days.
- Vary rolling window: test 500, 750, 1000 days.
- Vary spread z-score filter: test 0.3, 0.5, 0.8, 1.0, 1.5.
- Confirm that results are not driven by a small number of outlier trades.
- Test on multiple pairs simultaneously (GOOGL/META, JPM/BAC, XOM/CVX).

7. Daily Execution Workflow — Quick Reference

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Market Close (4:00 PM ET)
|
v
Compute log-returns r_A,t, r_B,t
|
v
Update GJR-GARCH -> get z_A,t, z_B,t, sigma_A,t, sigma_B,t
|
v
Update DCC -> get rho_t, h_t
|
v
PIT Transform -> get u_A,t, u_B,t
|
v
Compute Signal_t = dC(u_A,t, u_B,t) / du_A
Compute Spread Z_t = (S_t - mu_S) / sigma_S [NEW]
|
|-- Signal_t in (0.20, 0.80) -> NO ACTION
|
|-- Signal_t < 0.12 (1 consec. day)
|   AND rho_t > 0.40
|   AND |Z_t| > 0.3
|   AND no regime switch in last 5 days
|   AND currently flat
|       -> OPEN at next open: Long AMZN / Short MSFT
|           Notional per risk budget; hedge ratio h_t
|
|-- Signal_t > 0.88 (1 consec. day)
|   AND rho_t > 0.40
|   AND |Z_t| > 0.3
|   AND no regime switch in last 5 days
|   AND currently flat
|       -> OPEN at next open: Long MSFT / Short AMZN
|           Notional per risk budget; hedge ratio h_t
|
+-- IF POSITION IS OPEN:
|   |-- Signal_t enters [0.22, 0.78]
|   |   -> CLOSE 100% at next open (take-profit)
|   |
|   |-- Signal deepens into tail AND rho_t dropped > 0.15
|   |   -> CLOSE 100% at next open (corr-breakdown)
|   |
|   |-- Holding period >= 10 trading days
|   |   -> CLOSE 100% at next open (time stop)
|   |
|   |-- Intraday P&L loss > 1.8% NAV
|   |   -> CLOSE 100% immediately (hard stop)
|   |
|   +-- Hedge ratio drifted > 10%
|       -> REBALANCE short leg at next open
|
+-----+
| EVERY FRIDAY: Full Recalibration |
| (GJR-GARCH + DCC + Copula)       |
| + PIT validation + AIC test       |
| + Regime-change check             |
+-----+

```

8. Revision Log — Changes from Original Framework

#	Issue	Severity	Change Made
1	Position sizing had extra $\sqrt{252}$ in denominator, making positions ~16× too small	CRITICAL	Removed. Notional = (budget × NAV) / $\sigma_{\text{spread_annual}}$
2	Hedge ratio notation inconsistent (Long/Short changed meaning by direction)	HIGH	h_t always A/B; engine applies direction
3	Reversal protocol caused whipsaw losses on noisy signal jumps	HIGH	Removed entirely; exits via TP/stop only
4	Partial profit close split trades; second halves often became losers	HIGH	Replaced with single 100% close at [0.22, 0.78]
5	DCC Q_0 initialization unspecified	MEDIUM	$Q_0 = Q_{\text{bar}}$, re-initialized each recalibration
6	Signal divergence stop-loss criterion vague	MEDIUM	Quantified: signal deeper + ρ drop > 0.15
7	No secondary signal confirmation beyond copula	MEDIUM	Added spread z-score > 0.3 filter
8	GJR-GARCH convergence fallback absent	LOW	Multi-start optimization + EWMA fallback
9	BB1 copula had no stable h-function	LOW	Excluded; 4 families sufficient
10	Transaction cost timing not specified	LOW	Entry costs from NAV; exit costs from P&L;
11	Entry thresholds too strict for single pair	LOW	Relaxed to 0.12/0.88; confirmation reduced to 1 day
12	Time stop too loose (15 days)	LOW	Set to 10 days
13	Hard stop too loose (2% NAV) relative to avg win	LOW	Tightened to 1.8% NAV
14	Risk budget of 5% produced oversized positions after bug fix	LOW	Increased to 25%